

1 Exam Practice

Question 1.1 (Practice 1). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with compact support. Let $E \in \mathbb{R}$ and $\eta > 0$. Compute,

$$\lim_{\eta \rightarrow 0^+} \operatorname{Im} \left[\int_{\mathbb{R}} \frac{f(x)}{x - (E + i\eta)} dx \right]$$

If $\eta > 0$ show that the function

$$E \rightarrow \operatorname{Im} \left[\int_{\mathbb{R}} \frac{f(x)}{x - (E + i\eta)} dx \right]$$

is smooth.

Proof.

□

Question 1.2 (Practice 2). Show that a monotonic function $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable.

Proof.

□

Question 1.3 (Practice 3). If $m(E) = 0$ what can you say about $L^p(E)$?

Proof.

□

Question 1.4 (Practice 4). For which p are trigonometric polynomials dense in $L^p(-\pi, \pi)$?

Proof.

□

Question 1.5 (Practice 5). Let (M, d) be a metric space and for $x \in M$ and a nonempty set A define

$$d(x, A) := \inf_{y \in A} d(x, y)$$

Prove that $d(x, A)$ is a continuous function of x . Prove that A is closed iff the following statement holds: for all $x \in M$ we have that $d(x, A) = 0$ iff $x \in A$.

Proof.

□

Question 1.6 (Practice 6). Let $f_n(x)$ be a sequence of polynomials and let $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. Is f necessarily continuous?

Proof.

□

Question 1.7 (Practice 7). Let (M, d) be metric space. We say a sequence $x_n \in M$ is fast-Cauchy if there is a summable sequence $a_n > 0$ (i.e., $\sum a_n < \infty$) such that $d(x_n, x_{n+1}) < a_n$. Disprove or prove the following:

- (a) A fast-Cauchy sequence is Cauchy
- (b) Every Cauchy sequence has a fast-Cauchy subsequence
- (c) A sequence is Cauchy iff every subsequence has a subsequence that is fast-Cauchy

Proof.

□